# A. The Artful Expedient

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: brute force, implementation | https://codeforces.com/problemset/problem/869/A

*Rock... Paper!*

After Karen have found the deterministic winning (losing?) strategy for rock-paper-scissors, her brother, Koyomi, comes up with a new game as a substitute. The game works as follows.

A positive integer $n$ is decided first. Both Koyomi and Karen independently choose $n$ distinct positive integers, denoted by $x_1, x_2, ..., x_n$ and $y_1, y_2, ..., y_n$ respectively. They reveal their sequences, and repeat until **all of 2*n* integers become distinct**, which is the only final state to be kept and considered.

Then they count the number of ordered pairs $(i, j)$ $(1 \le i, j \le n)$ such that the value $x_i$ `xor` $y_j$ equals to one of the $2n$ integers. Here `xor` means the bitwise exclusive or operation on two integers, and is denoted by operators ^ and/or `xor` in most programming languages.

Karen claims a win if the number of such pairs is even, and Koyomi does otherwise. And you're here to help determine the winner of their latest game.

## Input

The first line of input contains a positive integer $n$ $(1 \le n \le 2\,000)$ — the length of both sequences.

The second line contains $n$ space-separated integers $x_1, x_2, ..., x_n$ $(1 \le x_i \le 2 \cdot 10^6)$ — the integers finally chosen by Koyomi.

The third line contains $n$ space-separated integers $y_1, y_2, ..., y_n$ $(1 \le y_i \le 2 \cdot 10^6)$ — the integers finally chosen by Karen.

Input guarantees that **the given 2*n* integers are pairwise distinct**, that is, no pair $(i, j)$ $(1 \le i, j \le n)$ exists such that one of the following holds: $x_i = y_j$; $i \ne j$ and $x_i = x_j$; $i \ne j$ and $y_i = y_j$.

## Output

Output one line — the name of the winner, that is, "`Koyomi`" or "`Karen`" (without quotes). Please be aware of the capitalization.

## Examples

| Input |
| --- |
| 3<br>1 2 3<br>4 5 6 |

| Output |
| --- |
| Karen |

| Input |
| --- |
| 5<br>2 4 6 8 10<br>9 7 5 3 1 |

| Output |
| --- |
| Karen |

## Note

In the first example, there are 6 pairs satisfying the constraint: (1, 1), (1, 2), (2, 1), (2, 3), (3, 2) and (3, 3). Thus, Karen wins since 6 is an even number.

In the second example, there are 16 such pairs, and Karen wins again.